

Calgary Residential Development ROI Classifier

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Motivation & Purpose

Real estate development is inherently risky, noisy, and unpredictable. Our system is an offline supervised-learning pipeline designed to classify completed Calgary residential developments into Low, Medium, or High Return on Investment (ROI). Its intended use is for research and early-stage deliberation, ranking projects for deeper expert review before significant capital is committed.

Disclaimer: This is not a deployed valuation service and must not be used as financial advice or to independently approve, reject, price, or finance a real development

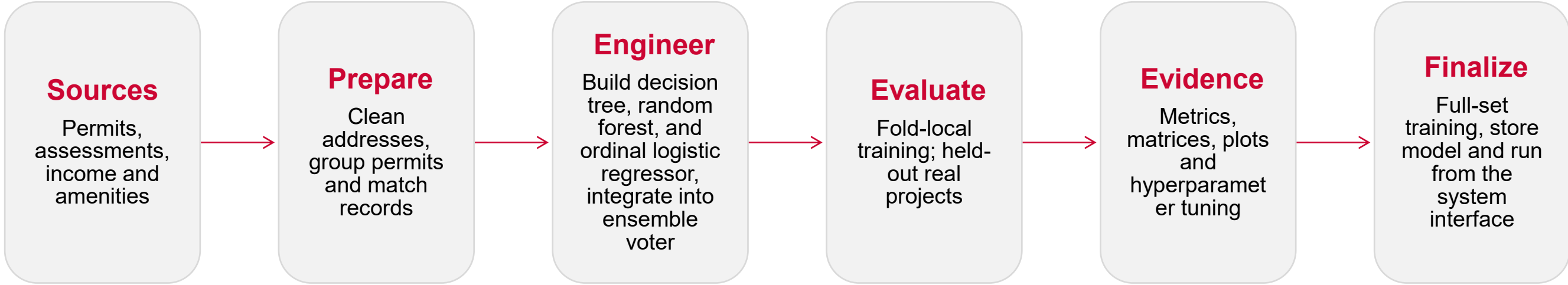
Project Overview & Target

Target: The model classifies ROI into three nearly balanced categories: 73 Low, 72 Medium, and 73 High.
Data Scope: Utilizing Calgary building permit applications (2017-2023) and completion records (2018-2023), paired with historical assessment records.
Inputs: A system interface takes four user inputs: a Calgary address, purchase price, and estimated number of units.
Feature Engineering: From the inputs, the system calculates community income, cost per unit, and distances to downtown, transit stops, parks, and public schools to pass to the classifier. Final assessment and ROI are strictly excluded from prediction features.

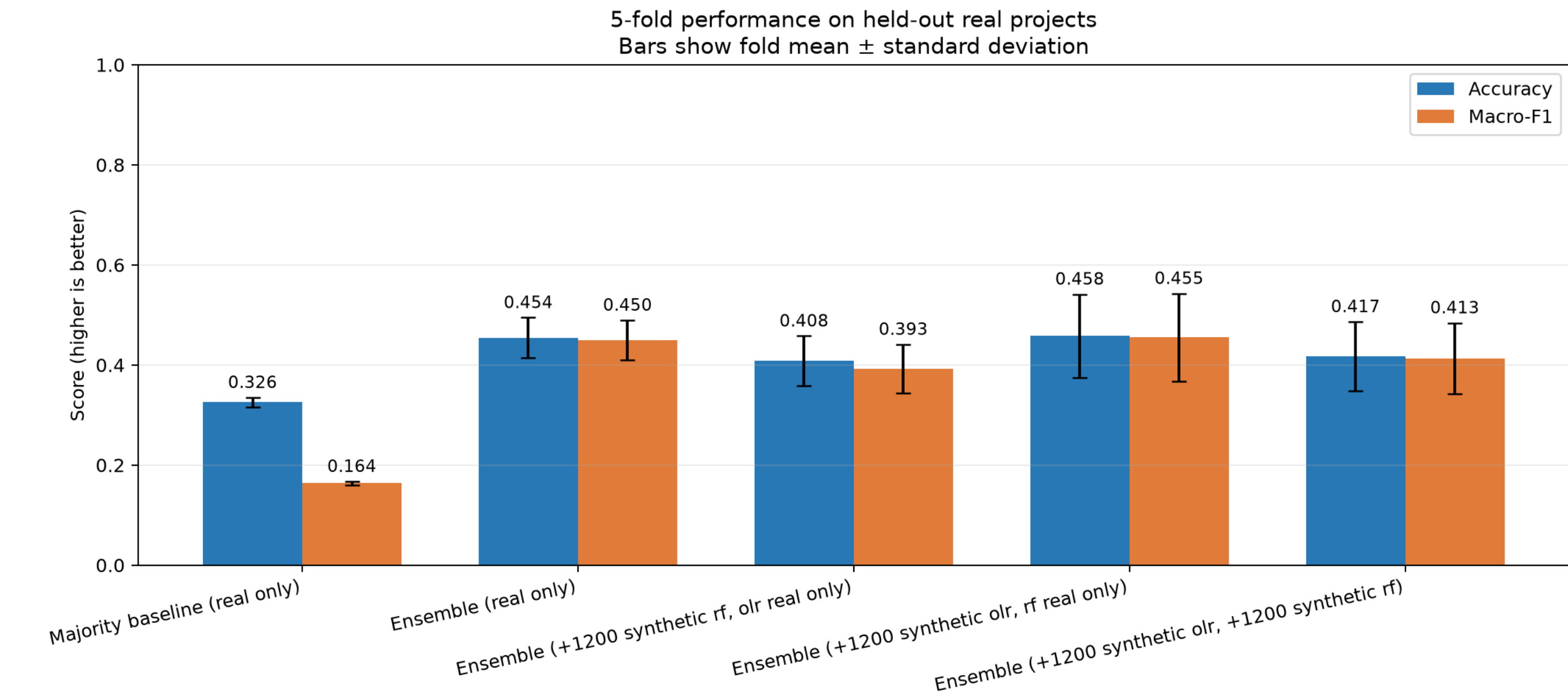
Methods & Technologies

Ensemble Modeling: The system utilizes a random forest classifier and an ordinal logistic regressor to form an ensemble voter using a soft voting system.
Evaluation Strategy: Evaluated using a 5-fold stratified cross-validation setup.
Leakage Prevention: New synthetic data is generated locally from each fold's training set to strictly ensure no data leakage.
Synthetic Data Generation: Attempted to mitigate data limitations by generating synthetic data following a multivariate, lognormal distribution.
Tech Stack: Python, NumPy, Pandas, Scikit-learn (for metrics, scaling, and baseline comparison), Haversine (distance calculations), and Flask (web interface).

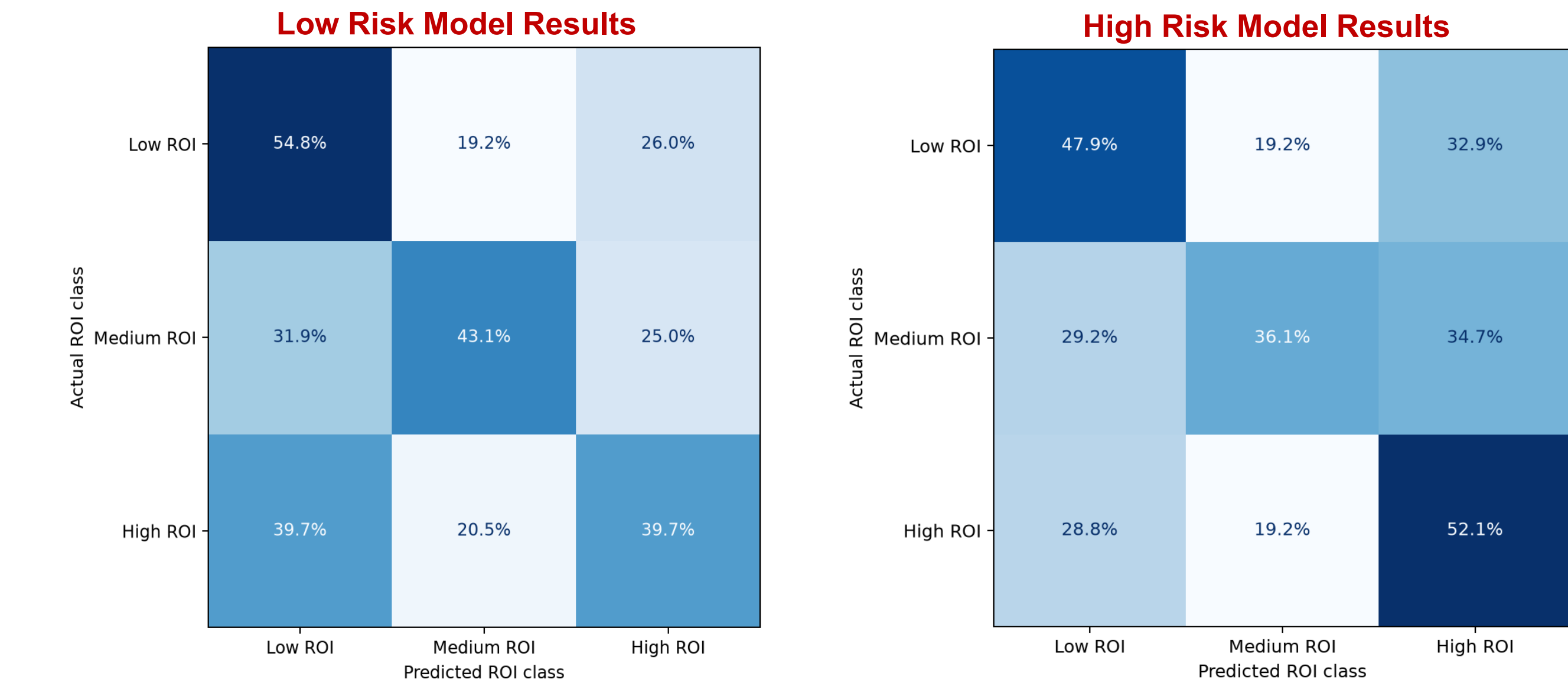
AI & Evaluation Pipeline



Results on Held-Out Real Projects



Risk Averse vs. Aggressive Investing Models



Key Results

Model	Training	Accuracy	Macro-F1	Ordinal MAE
Majority baseline	Real only	32.6% $\pm$ 0.9%	16.4 $\pm$ 0.3%	1.018
Ensemble	Real only	45.4% $\pm$ 4.0%	45.0% $\pm$ 4.0%	0.752
Ensemble	+1,200/fold for RF Real only for OLR	40.8% $\pm$ 5.0%	39.3% $\pm$ 4.9%	0.794
Ensemble	Real only for RF +1,200/fold for OLR	45.8% $\pm$ 8.3%	45.5% $\pm$ 8.8%	0.761
Ensemble	+1,200/fold for RF +1,200/fold for OLR	41.7% $\pm$ 6.9%	41.3% $\pm$ 7.1%	0.803

Result Conclusions

Highest Scorer: The real-only ensemble model achieved an accuracy of 45.4% and a Macro-F1 of 45.0%, significantly outperforming the majority class baseline (32.6% accuracy, 16.4% Macro-F1).

Synthetic Hypothesis: The synthetic-data hypothesis had mixed results: Adding 1,200 generated rows per fold into the ordinal logistic regression training resulted in the highest individual fold score (50%) but retained a greater variance/spread across accuracy and F1 metrics.

Model Trade-offs: The real-only ensemble acts as an "aggressive" predictor, balancing predictions between High and Low. The synthetic OLR model acts as a "lower-risk" model, heavily favoring "Low" predictions to avoid bad investments, thus reducing critical errors but increasing missed opportunities.

References

- City of Calgary Open Data datasets c2es-76ed, 4ur7-wsgc, muzh-c9qc, kami-qbfh, and fd9t-tdn2.
- Canada Mortgage and Housing Corporation, Housing Market Information Portal.
- Group 14 project repository and generated evidence under Artifacts/Milestone2.